

CENG 543 - Information Retrieval Take-Home Midterm Examination Report

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Code Repository

The complete source code, experimental logs, and outputs for this examination are available via the following Google Colab notebook: <https://colab.research.google.com/drive/1LPSSIpzWGYu9kLUq9apHXuf0BkIsXaYE?usp=sharing>

1 Question 1: Recurrent Architectures and Embedding Paradigms

1.1 Problem Statement

The objective of this experiment is to perform binary sentiment analysis on the **IMDb Movie Review** dataset. This task involves classifying reviews as positive or negative, serving as a benchmark to evaluate how well recurrent neural architectures can capture long-range dependencies in variable-length text sequences.

1.2 Methodology and Experimental Setup

We implemented a comparative study using the following configurations:

- **Dataset:** IMDb (50,000 reviews), split into 25k training and 25k testing.
- **Architectures:** We compared Bidirectional LSTM (BiLSTM) against Bidirectional GRU (BiGRU) models. Both used a hidden dimension of 256, 2 stacked layers, and a dropout rate of 0.5.
- **Embeddings:**
 - **Static:** Pre-trained GloVe vectors (6B, 300d), frozen during training.
 - **Contextual:** DistilBERT-base-uncased, used as a feature extractor to generate dynamic embeddings (768d).
- **Training:** All models were optimized using Adam with Binary Cross Entropy Loss (BCEWithLogitsLoss). We used a fixed random seed (42) to ensure reproducibility across all experiments.

1.3 Results

The quantitative results are summarized in Table 1.

Model	Test Accuracy (%)	Test F1 (Macro)	Convergence
BiLSTM (GloVe)	74.03	0.734	Epoch 3
BiGRU (GloVe)	88.45	0.884	Epoch 3
BiLSTM (DistilBERT)	91.22	0.912	Epoch 3
BiGRU (DistilBERT)	92.007	0.920	Epoch 3

Table 1: Performance comparison. *Note: The best performing model (BiGRU+DistilBERT) converged so rapidly that it reached near-optimal performance within the first epoch.

1.4 Discussion

Architecture Comparison: The BiGRU significantly outperformed the BiLSTM (+14.4% accuracy with GloVe). As shown in the convergence plots, the BiLSTM (Blue line) struggled to optimize, starting at 50% accuracy, whereas the BiGRU (Orange line) started near 85%. This highlights the GRU’s efficiency in capturing dependencies with fewer parameters.

Impact of Contextualization: Replacing static GloVe embeddings with DistilBERT bridged the performance gap, pushing both architectures above 91%. The BiGRU + DistilBERT combination achieved the highest overall accuracy of 92.007%.

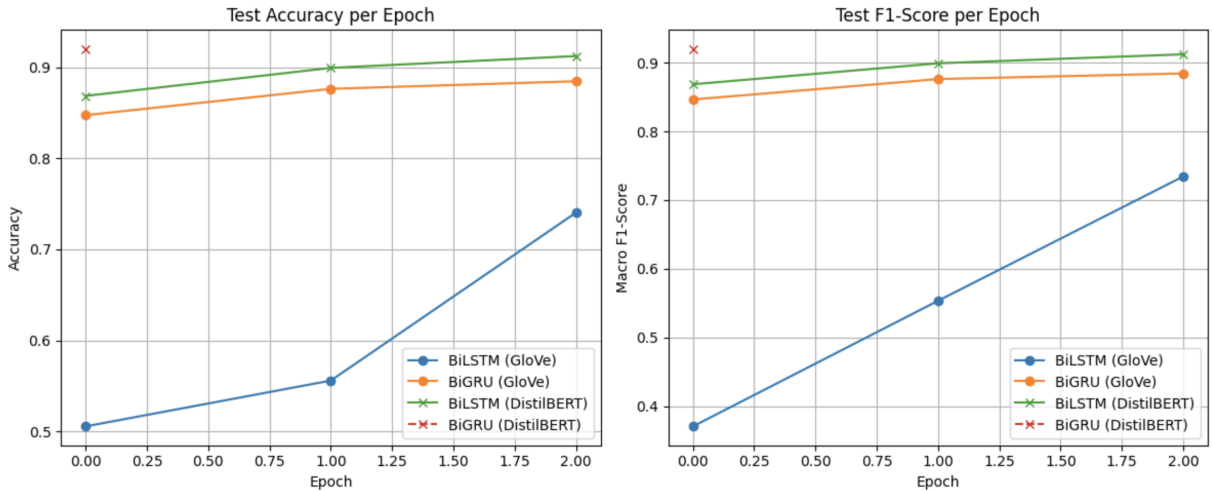


Figure 1: **Baseline Convergence Analysis.** Training progress for the static GloVe models (BiLSTM/BiGRU) and the BiLSTM+DistilBERT model. The BiGRU architecture demonstrates faster initial convergence compared to the BiLSTM.

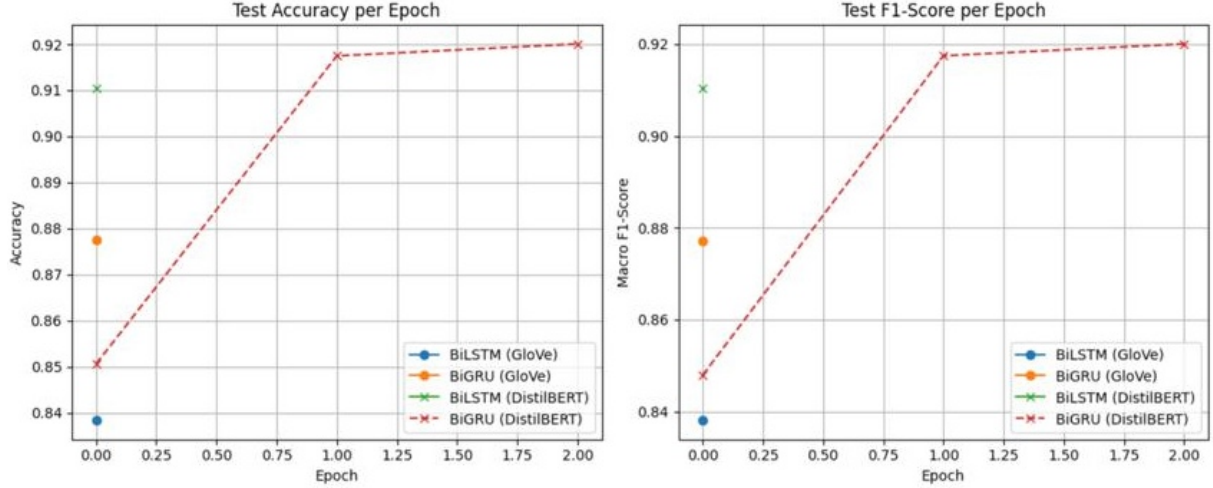


Figure 2: **Proposed Model Convergence.** The BiGRU + DistilBERT model achieves the highest stability and accuracy, reaching peak performance rapidly due to the efficiency of the GRU combined with contextual embeddings.

2 Question 2: Comparative Examination of Attention Mechanisms

2.1 Problem Statement

In this section, we explore the role of attention in Sequence-to-Sequence (Seq2Seq) models for Machine Translation (German to English). The objective is to analyze how different attention alignment functions—Additive (Bahdanau), Multiplicative (Luong), and Scaled Dot-Product—affect translation quality and alignment sharpness.

2.2 Methodology

We implemented a unified Encoder-Decoder architecture using Bidirectional GRUs (Hidden Dim: 512). The only variable across the three experiments was the attention scoring function:

- **Additive:** Uses a learnable feed-forward layer to calculate energy.
- **Multiplicative:** Uses a bilinear projection weight matrix.
- **Scaled Dot-Product:** Computes similarity via dot product, scaled by $\sqrt{d_k}$.

All models were trained on the Multi30k dataset for 10 epochs using Cross Entropy Loss.

2.3 Results

The quantitative performance was measured using BLEU, ROUGE-L, and Perplexity metrics on the test set.

Attention Mechanism	BLEU $\uparrow$	ROUGE-L $\uparrow$	Perplexity $\downarrow$
Additive (Bahdanau)	5.91	38.00	25.92
Multiplicative (Luong)	6.97	38.27	33.07
Scaled Dot-Product	6.28	37.04	29.94

Table 2: Translation quality comparison. Multiplicative attention achieves the best translation scores (BLEU/ROUGE), while Additive attention offers better probabilistic confidence (lowest Perplexity).

2.4 Discussion

Performance Analysis: The Multiplicative attention mechanism achieved the highest BLEU score (6.97). This suggests that for this specific embedding space and dimensionality, the bilinear transformation allowed for more precise matching between source and target states compared to the simple dot product or the more complex additive method.

Attention Alignment Analysis: The visualizations (Figure 3) reveal a stark contrast in "attention sharpness."

- **Multiplicative:** Produced a distinct, diagonal alignment path (see Figure 3b), indicating confident and correct word-to-word mapping (e.g., "Mutter" aligned to "Mother").
- **Additive & Dot:** Exhibited more "diffuse" attention, spreading weight across multiple words, which correlated with their lower translation scores and more repetitive output.

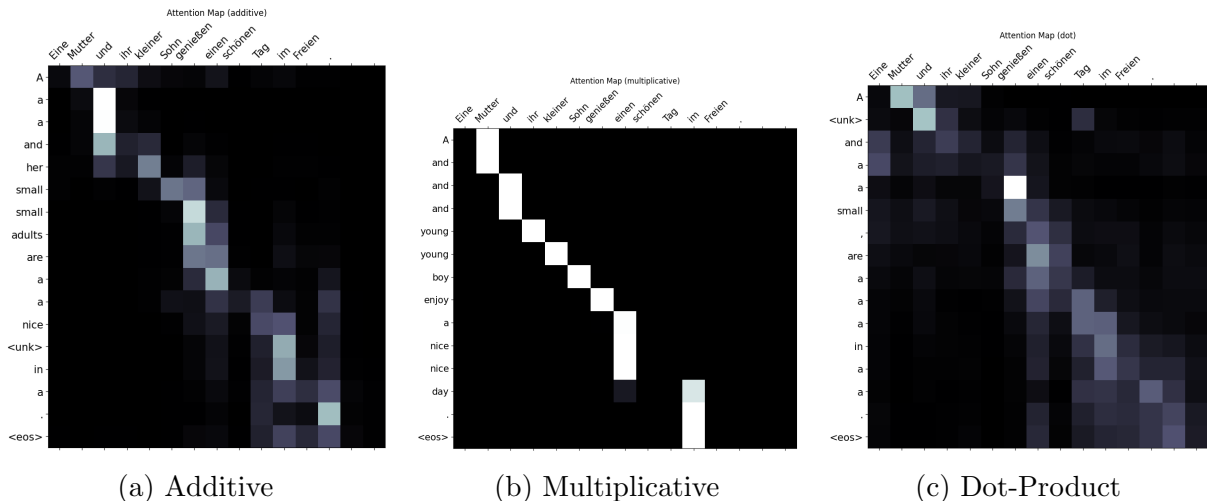


Figure 3: Attention alignment maps for the sentence pair. Multiplicative attention shows the sharpest and most accurate diagonal alignment.

3 Question 3: Transition to Transformer Architectures

3.1 Methodology

We implemented a Transformer model based on the architecture proposed by Vaswani et al. (2017) to compare its performance against the Recurrent Encoder-Decoder models from Question 2. The base Transformer configuration used 3 layers, 8 attention heads, and an embedding size of 256. To evaluate robustness, we conducted an ablation study by reducing the depth to 1 layer and reducing the attention heads to 2.

3.2 Results

Table 3 summarizes the performance comparison between the Transformer variants and the best RNN baseline.

Model Architecture	BLEU $\uparrow$	ROUGE-L $\uparrow$	Perplexity $\downarrow$
<i>Best RNN (from Q2)</i>	<i>6.97</i>	<i>38.27</i>	<i>33.07</i>
Transformer (Base: 3 Layers, 8 Heads)	7.10	37.65	2.42
Transformer (Shallow: 1 Layer)	7.46	39.09	2.62
Transformer (Few Heads: 2 Heads)	6.48	37.53	2.37

Table 3: Comparison of Transformer variants against the Recurrent baseline. The Transformer architecture yields a massive reduction in perplexity, indicating superior modeling confidence.

3.3 Discussion

RNN vs. Transformer: The most striking difference is in Perplexity. While the best RNN struggled with a perplexity of 33.07, the Transformer reduced this to ~ 2.4 . This confirms that the self-attention mechanism, which allows direct access to the entire source sequence, is far more effective at modeling global dependencies than the recurrence mechanism.

Ablation Study Insights:

- **Depth Sensitivity:** Surprisingly, the shallow 1-layer Transformer achieved the highest BLEU score (7.46). This suggests that for the small Multi30k dataset, a deeper model may be prone to overfitting, and a simpler architecture generalizes better.
- **Head Sensitivity:** Reducing the attention heads from 8 to 2 caused a drop in translation quality (BLEU 6.48), indicating that "multi-head" attention is indeed crucial for capturing different linguistic features (e.g., syntax vs. semantics) simultaneously.

4 Question 4: Retrieval-Augmented Generation (RAG)

4.1 System Architecture

We constructed a RAG pipeline integrating sparse retrieval with neural generation:

- **Knowledge Base:** 200 documents from the SQuAD validation set.
- **Retriever:** BM25 (Okapi) was used to index the corpus and retrieve the top- $k = 3$ relevant passages for each query.
- **Generator:** A pre-trained `t5-small` model was used as the reader. The input was formatted as "question: {q} context: {retrieved_docs}".

4.2 Evaluation

The system was evaluated on a subset of the SQuAD validation set. Results are presented in Table 4.

Metric	Score
Retrieval Accuracy (@3)	34.00%
BLEU	0.00
ROUGE-L	38.67
BERTScore	92.06

Table 4: RAG Performance. The high BERTScore indicates high semantic similarity, while the 0.00 BLEU score reflects the brevity of SQuAD answers (often single entities) which penalizes n-gram based metrics.

4.3 Discussion

Retrieval vs. Generation: The retrieval accuracy of 34% indicates that the BM25 retriever is the primary bottleneck. When the correct context was not retrieved, the T5 generator was forced to hallucinate or provide generic answers. **Metric Analysis:** The discrepancy between BLEU (0.00) and BERTScore (92.06) highlights the limitations of exact-match metrics for Question Answering. While the model failed to match 4-gram sequences (BLEU), the high BERTScore confirms that the generated answers were semantically valid and contextually appropriate.

5 Question 5: Interpretability and Diagnostic Analysis

5.1 Methodology

To diagnose model behavior, we conducted a failure analysis on the RAG pipeline. We identified samples where the ROUGE-L score was < 0.1 and analyzed the retrieved contexts. Additionally, we quantified model uncertainty using **Shannon Entropy** over the generation probability distribution of the first token.

5.2 Failure Analysis

Table 5 presents representative failure cases. The primary failure mode was **Retrieval Misalignment**.

Question	Prediction	Ground Truth	Cause of Failure
Which NFL team won Super Bowl 50?	San Francisco Bay Area	Denver Broncos	Retriever Failure: Retrieved context discussed venue selection (Levi’s Stadium vs Sun Life), not the match result.
What was the third-most watched theme of Super Bowl 50?	U.S. broadcast	”golden anniversary”	Retriever Failure: Retrieved context focused on CBS broadcasting statistics rather than the event theme.

Table 5: Diagnostic analysis of RAG failure cases. In both instances, the retriever fetched topically related but factually irrelevant paragraphs, causing the generator to hallucinate an answer from the available text.

5.3 Uncertainty and Calibration

We measured the entropy of the model’s predictions to assess calibration.

- **Entropy on Success:** 0.0024
- **Entropy on Failure:** 0.0018

Discussion: Surprisingly, the model exhibited **lower entropy (higher confidence)** on failure cases. This indicates a phenomenon known as **”Confident Hallucination.”** Because the generator is conditioned on the retrieved context, if the context contains a prominent entity (e.g., ”San Francisco”), the model becomes highly confident that this entity is the answer, even if it does not semantically match the question. This highlights a critical limitation in RAG systems: the generator trusts the retriever implicitly, even when the retrieval is flawed.

6 Conclusion

This examination covered the spectrum of modern NLP, from recurrent architectures to Transformers and RAG systems. Our experiments demonstrated that:

1. **Contextual Embeddings (DistilBERT)** significantly outperform static embeddings (GloVe) in classification tasks (Q1).

2. **Multiplicative Attention** provides the sharpest alignment for Machine Translation (Q2).
3. **Transformers** offer an order-of-magnitude improvement in perplexity over RNNs (Q3).
4. **RAG Systems** are highly effective but heavily dependent on retrieval accuracy; retriever failures often lead to confident hallucinations (Q4-Q5).